2.4 Analysis in $\mathbb{R}^n$: Study Notes

Notation: $|x|$ is the Euclidean norm on $\mathbb{R}^n$; for a linear map $T:\mathbb{R}^n\to\mathbb{R}^m$, $\|T\|=\sup_{|x|\le 1}|Tx|$ is the operator norm. K denotes a compact metric space and $C(K)$ the continuous real functions on K with $\|f\|_\infty=\sup_K|f|$.

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1. Sequences and series of functions

Definitions. $f_n\to f$ *pointwise* on E if $f_n(x)\to f(x)$ for each $x\in E$. $f_n\to f$ *uniformly* if $\sup_{x\in E}|f_n(x)-f(x)|\to 0$. Equivalently (Cauchy criterion), for every ε there is N with $|f_n(x)-f_m(x)|<\varepsilon$ for all x and all $m,n\ge N$.

Theorems.

1. *Continuity.* A uniform limit of continuous functions is continuous. ($\varepsilon/3$: $|f(x)-f(y)|\le |f(x)-f_N(x)|+|f_N(x)-f_N(y)|+|f_N(y)-f(y)|$.)
2. *Completeness.* $(C(K),\|\cdot\|_\infty)$ is a complete normed space (a Banach space).
3. *Integrals.* If $f_n\to f$ uniformly on $[a,b]$ and the f_n are Riemann integrable, then f is Riemann integrable and $\int_a^b f_n\to\int_a^b f$.
4. *Derivatives.* If f_n are differentiable on $[a,b]$, $(f_n(x_0))$ converges for some x_0, and (f_n') converges **uniformly**, then f_n converges uniformly to some f and $f'=\lim f_n'$. (Uniform convergence of f_n alone is not enough: $f_n(x)=\frac{\sin(nx)}{\sqrt{n}}\to 0$ uniformly, but $f_n'(0)=\sqrt{n}\to\infty$.)
5. *Weierstrass M-test.* If $|g_k(x)|\le M_k$ on E and $\sum M_k<\infty$, then $\sum g_k$ converges uniformly (and absolutely) on E.
6. *Dini's theorem.* If $f_n\in C(K)$ increase pointwise to a continuous f, the convergence is uniform.

Standard counterexamples.

- x^n on $[0,1]$: continuous functions with a discontinuous pointwise limit.
- nxe^{-nx^2} on $[0,1]$: pointwise limit 0, but $\int_0^1 nxe^{-nx^2}dx\to\frac{1}{2}$.
- Tent functions of height n and width $2/n$: pointwise to 0, integrals equal to 1.
- $\sum\frac{\sin nx}{n^2}$ converges uniformly, but the differentiated series $\sum\frac{\cos nx}{n}$ diverges at $x=0$.

Power series. $\sum a_nx^n$ with radius R converges uniformly on $[-r,r]$ for every $r<R$ (M-test), can be integrated and differentiated term by term inside $(-R,R)$.

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2. Arzelà–Ascoli

Definition. A family $\mathcal{F}\subseteq C(K)$ is *equicontinuous* if for every $\varepsilon>0$ there is $\delta>0$ such that $d(x,y)<\delta\Rightarrow|f(x)-f(y)|<\varepsilon$ for **all** $f\in\mathcal{F}$ (one δ for the whole family, and uniform in x). It is *uniformly bounded* if $\sup_{f\in\mathcal{F}}\|f\|_\infty<\infty$.

Arzelà–Ascoli theorem. Let K be a compact metric space and $\mathcal{F}\subseteq C(K)$. Then $\mathcal{F}$ has compact closure in $C(K)$ (equivalently, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence) iff $\mathcal{F}$ is uniformly bounded and equicontinuous. (Pointwise boundedness suffices in place of uniform boundedness, given equicontinuity and compactness of K.)

Proof of "if". Let (f_n) be a sequence in $\mathcal{F}$.

1. A compact metric space has a countable dense subset $D = \{x_1, x_2, \dots\}$ (take centres of finite $1/k$-nets for each k).
2. *Diagonal argument.* $(f_n(x_1))$ is a bounded sequence of reals, so has a convergent subsequence. From it extract a subsequence converging at x_2, and so on. The diagonal sequence g_k (the k-th term of the k-th subsequence) converges at every x_j.
3. *Upgrade to uniform convergence.* Given ε, take δ from equicontinuity. By compactness, finitely many points $x_{j_1}, \dots, x_{j_r}$ of D form a δ-net. Choose N such that $|g_k(x_{j_i}) - g_l(x_{j_i})| < \varepsilon$ for $k, l \geq N$ and all $i \leq r$. For any x, pick x_{j_i} within δ; then

$$|g_k(x) - g_l(x)| \leq |g_k(x) - g_k(x_{j_i})| + |g_k(x_{j_i}) - g_l(x_{j_i})| + |g_l(x_{j_i}) - g_l(x)| < 3\varepsilon.$$

So (g_k) is uniformly Cauchy, hence uniformly convergent by completeness of $C(K)$. □

"Only if" is a total-boundedness argument: a compact set in $C(K)$ is covered by finitely many ε-balls, and finitely many continuous functions on a compact space are uniformly equicontinuous.

Typical applications.

- $\{f \in C^1[0,1] : |f(0)| \leq 1, |f'| \leq 1\}$ has compact closure in $C[0,1]$ (Lipschitz with constant 1 gives equicontinuity).
- Integral operators with continuous kernels map bounded sets to relatively compact sets (*compact operators*, module 3.2).
- *Peano's existence theorem* for $y' = F(t, y)$ with F only continuous (Euler polygons, then Arzelà–Ascoli).
- *Montel's theorem* in complex analysis (module 2.2).

Pitfall. $\{\sin(nx)\}$ on $[0, \pi]$ is uniformly bounded but not equicontinuous, and has no uniformly convergent subsequence.

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3. Stone–Weierstrass

Weierstrass approximation theorem. Polynomials are dense in $C[a,b]$. (One constructive proof uses Bernstein polynomials $B_n f(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k} x^k (1-x)^{n-k}$ on $[0,1]$; another convolves with polynomial kernels $c_n(1-x^2)^n$.)

Stone–Weierstrass theorem (real). Let K be compact Hausdorff (for us, compact metric) and $\mathcal{A} \subseteq C(K, \mathbb{R})$ a subalgebra that contains the constants and *separates points* (for $x \neq y$ there is $f \in \mathcal{A}$ with $f(x) \neq f(y)$). Then $\mathcal{A}$ is dense in $C(K)$.

Proof sketch.

1. The closure $\overline{\mathcal{A}}$ is again an algebra.
2. If $f \in \overline{\mathcal{A}}$ then $|f| \in \overline{\mathcal{A}}$: by Weierstrass, $|t|$ is a uniform limit of polynomials $p_n(t)$ on $[-\|f\|, \|f\|]$, and $p_n \circ f \in \overline{\mathcal{A}}$. Hence $\max(f,g) = \frac{f+g+|f-g|}{2}$ and $\min(f,g)$ lie in $\overline{\mathcal{A}}$.
3. For $x \neq y$ and any reals α, β there is $h \in \mathcal{A}$ with $h(x) = \alpha$, $h(y) = \beta$ (take $h = \alpha + (\beta - \alpha)\frac{u - u(x)}{u(y) - u(x)}$ with u separating x, y).
4. Fix $f \in C(K)$, $\varepsilon > 0$. For fixed x and each y choose $h_y \in \mathcal{A}$ with $h_y(x) = f(x)$, $h_y(y) = f(y)$. The open sets $\{h_y < f + \varepsilon\}$ cover K; take finitely many and let $g_x = \min_i h_{y_i} \in \overline{\mathcal{A}}$. Then $g_x < f + \varepsilon$ everywhere and $g_x(x) = f(x)$.
5. The open sets $\{g_x > f - \varepsilon\}$ cover K; take finitely many and let $g = \max_j g_{x_j} \in \overline{\mathcal{A}}$. Then $f - \varepsilon < g < f + \varepsilon$. □

Complex version. For $C(K,\mathbb{C})$, also require $\mathcal{A}$ to be closed under complex conjugation. Without it the theorem fails: polynomials in z are not dense in $C(\overline{\mathbb{D}},\mathbb{C})$, since uniform limits are holomorphic inside the disc.

Applications.

- Trigonometric polynomials are dense in $C(S^1)$ (continuous 2π-periodic functions): the algebra generated by $e^{\pm i\theta}$ is conjugation-closed and separates points of the circle.
- Functions $\sum g_i(x)h_i(y)$ are dense in $C([0,1]^2)$.
- *Moment problem:* if $f \in C[0,1]$ and $\int_0^1 f x^n = 0$ for all n, then $f = 0$.
- Polynomials in x^2 are dense in $C[0,1]$ but not in $C[-1,1]$ (they do not separate $\pm x$).

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4. Contraction mapping and ODEs

Contraction mapping theorem (see module 2.1). A contraction of a non-empty complete metric space has a unique fixed point, the limit of the iterates from any starting point.

Picard–Lindelöf. Let $F : [t_0 - a, t_0 + a] \times \mathbb{R}^n \to \mathbb{R}^n$ be continuous and Lipschitz in y: $|F(t,y) - F(t,z)| \le L|y - z|$. Then $y' = F(t,y)$, $y(t_0) = y_0$ has a unique solution on $[t_0 - h, t_0 + h]$ for any $h < \min(a, 1/L)$.

Proof. A C^1 solution is the same as a continuous solution of the integral equation $y(t) = y_0 + \int_{t_0}^t F(s, y(s))\,ds$ (fundamental theorem of calculus). The map $(Ty)(t) = y_0 + \int_{t_0}^t F(s, y(s))\,ds$ sends $C([t_0 - h, t_0 + h], \mathbb{R}^n)$ to itself and $\|Ty - Tz\|_\infty \le hL\|y - z\|_\infty$, a contraction. (With only local Lipschitz bounds, restrict to a closed ball around the constant function y_0, which is still complete.) □

The Lipschitz condition matters: $y' = 3y^{2/3}$, $y(0) = 0$ has the two solutions $y = 0$ and $y = t^3$.

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5. The derivative as a linear map

Definition. $f : U \to \mathbb{R}^m$ ($U \subseteq \mathbb{R}^n$ open) is *differentiable* at a if there is a linear $T = Df(a)$ with

$$\lim_{h \to 0} \frac{|f(a+h) - f(a) - Th|}{|h|} = 0.$$

T is unique; its matrix is the Jacobian $\left(\partial f_i / \partial x_j(a)\right)$.

Facts.

1. Differentiable ⇒ continuous; differentiable ⇒ all directional derivatives exist, with $D_v f(a) = Df(a)v$.
2. The converse fails: $f(x,y) = \frac{xy}{x^2+y^2}$ ($f(0) = 0$) has both partial derivatives at 0 but is not continuous there. $f(x,y) = \frac{x^2 y}{x^4+y^2}$ has all directional derivatives at 0 but is not continuous there.
3. **If all partial derivatives exist near a and are continuous at a, then f is differentiable at a.** (Move one coordinate at a time and apply the 1-variable mean value theorem.) f is C^1 iff its partials exist and are continuous iff $x \mapsto Df(x)$ is continuous into the space of linear maps.
4. **Chain rule:** $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$.

5. **Mean value inequality:** if U is convex and $\|Df(x)\| \le M$ on U, then $|f(x)-f(y)| \le M|x-y|$. (Apply the 1-variable MVT to $t \mapsto \langle f(x+t(y-x))-f(x), u\rangle$ for a suitable unit vector u.) There is no mean value *equality* for vector-valued f: $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi]$.

Examples worth knowing.

- $D(A \mapsto A^2)(A)H = AH + HA$.
- $D(A \mapsto A^{-1})(A)H = -A^{-1}HA^{-1}$ on the open set $GL_n(\mathbb{R})$.
- $D(\det)(I)H = \operatorname{tr} H$, and $D(\det)(A)H = \det(A)\operatorname{tr}(A^{-1}H)$ for invertible A.
- A bilinear map B has $DB(x,y)(h,k) = B(h,y) + B(x,k)$ (product rule).

Higher derivatives. $D^2 f(a)$ is a bilinear map, with matrix the Hessian. If f is C^2, it is symmetric ($\partial_i\partial_j f = \partial_j\partial_i f$). Without continuity of second partials this can fail: $f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0)=0$, has $\partial_x\partial_y f(0) \ne \partial_y\partial_x f(0)$. **Taylor's theorem:** $f(a+h) = f(a) + Df(a)h + \frac{1}{2}D^2 f(a)(h,h) + o(|h|^2)$ for C^2 f. At a critical point, positive definite Hessian ⇒ strict local minimum.

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6. The inverse function theorem

Theorem. Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}^n$ of class C^1, $a \in U$, and $Df(a)$ invertible. Then there are open sets $V \ni a$ and $W \ni f(a)$ such that $f : V \to W$ is a bijection, and its inverse $g : W \to V$ is C^1 with $Dg(y) = Df(g(y))^{-1}$. If f is C^k, so is g.

Proof. Let $A = Df(a)$ and choose λ with $2\lambda\|A^{-1}\| = 1$. By continuity of Df, choose an open ball V around a with $\|Df(x) - A\| < \lambda$ for $x \in V$.

Step 1 (injectivity). For $y \in \mathbb{R}^n$ define $\varphi_y(x) = x + A^{-1}(y - f(x))$. Then $f(x) = y$ iff x is a fixed point of φ_y. Since $D\varphi_y(x) = A^{-1}(A - Df(x))$ has norm $< \|A^{-1}\|\lambda = \frac{1}{2}$ on the convex set V, the mean value inequality gives $|\varphi_y(x_1) - \varphi_y(x_2)| \le \frac{1}{2}|x_1 - x_2|$. A map that shrinks distances has at most one fixed point, so f is injective on V.

Step 2 ($W = f(V)$ is open). Let $y_0 = f(x_0)$, $x_0 \in V$, and choose $r > 0$ with $\overline{B}(x_0, r) \subseteq V$. If $|y - y_0| < \lambda r$ then for $x \in \overline{B}(x_0, r)$,

$$|\varphi_y(x) - x_0| \le |\varphi_y(x) - \varphi_y(x_0)| + |\varphi_y(x_0) - x_0| \le \tfrac{1}{2}r + \|A^{-1}\|\,|y - y_0| < \tfrac{r}{2} + \tfrac{r}{2} = r.$$

So φ_y maps the complete space $\overline{B}(x_0, r)$ into itself and is a contraction; its fixed point x satisfies $f(x) = y$. Hence $B(y_0, \lambda r) \subseteq W$.

Step 3 (g is differentiable). Note that $\|Df(x) - A\| < \lambda$ and $\|A^{-1}\|\lambda = \frac{1}{2}$ imply $Df(x)$ is invertible on V. Let $y, y+k \in W$, $x = g(y)$, $h = g(y+k) - x$. From Step 1, $|h - A^{-1}k| = |\varphi_y(x+h) - \varphi_y(x)| \le \frac{1}{2}|h|$, so $|h| \le 2\|A^{-1}\|\,|k| = |k|/\lambda$. With $T = Df(x)^{-1}$,

$$g(y+k) - g(y) - Tk = h - Tk = -T\big(f(x+h) - f(x) - Df(x)h\big),$$

and dividing by $|k| \ge \lambda|h|$ shows this is $o(|k|)$. So $Dg(y) = Df(g(y))^{-1}$. Since g is continuous, Df is continuous, and matrix inversion is continuous (indeed smooth: Cramer's rule), Dg is continuous. Bootstrapping gives C^k. □

Pitfalls.

- The theorem is local. $f(x,y) = (e^x \cos y, e^x \sin y)$ has invertible derivative everywhere but is not injective on $\mathbb{R}^2$.

- C^1 matters: $f(x) = x + 2x^2 \sin(1/x)$ ($f(0) = 0$) has $f'(0) = 1$ but is not injective on any neighbourhood of 0.
- $Df(a)$ invertible is sufficient, not necessary, for a continuous inverse ($x \mapsto x^3$), but it is necessary for a *differentiable* inverse.

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7. The implicit function theorem and consequences

Implicit function theorem. Let $F : U \to \mathbb{R}^m$ be C^1 on an open $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$, with $F(a,b) = 0$ and $D_yF(a,b)$ (the $m \times m$ block of partials in y) invertible. Then there are open $V \ni a$, $W \ni b$ and a unique C^1 map $g : V \to W$ such that, for $(x,y) \in V \times W$, $F(x,y) = 0 \iff y = g(x)$. Its derivative is

$$Dg(x) = -D_yF(x,g(x))^{-1}\, D_xF(x,g(x)).$$

Proof. Apply the inverse function theorem to $\Phi(x,y) = (x, F(x,y))$, whose derivative $\begin{pmatrix} I & 0 \\ D_xF & D_yF \end{pmatrix}$ is invertible at (a,b). Then $g(x)$ is the y-component of $\Phi^{-1}(x,0)$. The derivative formula follows by differentiating $F(x,g(x)) = 0$ with the chain rule. □

Rank theorem (statement). If f is C^1 and Df has constant rank r near a, then in suitable local C^1 coordinates around a and $f(a)$, $f(x_1,\dots,x_n) = (x_1,\dots,x_r,0,\dots,0)$. Special cases: submersions (rank m) look like projections; immersions (rank n) look like inclusions.

Regular level sets. If $F : \mathbb{R}^n \to \mathbb{R}^m$ is C^1 and $DF(p)$ is surjective at every p with $F(p) = c$, then $F^{-1}(c)$ is locally the graph of a C^1 function of $n-m$ variables: an $(n-m)$-dimensional submanifold (module 3.5). Example: $S^{n-1} = \{|x|^2 = 1\}$.

Lagrange multipliers. If $f, g_1, \dots, g_m$ are C^1, p is a local extremum of f on $M = \{g = 0\}$, and $Dg(p)$ is surjective, then $\nabla f(p) = \sum \lambda_i \nabla g_i(p)$. (The tangent space $\ker Dg(p)$ consists of velocities of curves in M through p, by the implicit function theorem, and $Df(p)$ must vanish on it.)

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8. Riemann integration in $\mathbb{R}^n$

Definition. For a closed rectangle $R = \prod[a_i, b_i]$ and bounded $f : R \to \mathbb{R}$, partitions of R into subrectangles give upper and lower sums $U(f,P)$, $L(f,P)$. f is *Riemann integrable* if $\sup_P L(f,P) = \inf_P U(f,P)$, the common value being $\int_R f$.

Lebesgue's criterion (statement). A bounded f on R is Riemann integrable iff its set of discontinuities has (Lebesgue) measure zero, i.e. can be covered by countably many rectangles of arbitrarily small total volume.

Jordan measurable sets. A bounded set E is *Jordan measurable* if ∂E has measure zero (equivalently, the indicator function χ_E is Riemann integrable). For such E and bounded f continuous except on a null set, $\int_E f := \int_R f\chi_E$ for any rectangle $R \supseteq E$. Balls, polytopes and regions bounded by finitely many C^1 graphs are Jordan measurable. $\mathbb{Q} \cap [0,1]$ is not (its boundary is $[0,1]$).

Fubini's theorem (Riemann version). Let f be integrable on $A \times B$ ($A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^m$ rectangles). Then the lower and upper integrals $\underline{\int}_B f(x,\cdot)$ and $\overline{\int}_B f(x,\cdot)$ are integrable functions

of x on A, and

$$\int_{A\times B} f = \int_A \Big(\underline{\int_B} f(x,y)\,dy\Big)dx = \int_A \Big(\overline{\int_B} f(x,y)\,dy\Big)dx.$$

In particular, for continuous f, $\int_{A\times B} f = \int_A \int_B f\,dy\,dx = \int_B \int_A f\,dx\,dy$.

Proof idea. For a partition $P = P_A \times P_B$, $L(f,P) \le L(\underline{\int}_B f(x,\cdot), P_A) \le U(\overline{\int}_B f(x,\cdot), P_A) \le U(f,P)$, and the outer two converge to $\int f$.

Pitfalls.

- Iterated integrals can exist and differ when f is not integrable on the product: $f(x,y) = \frac{x^2-y^2}{(x^2+y^2)^2}$ on $(0,1]^2$ (Problem 17).
- Iterated integrals can exist and agree while f is not Riemann integrable on the square (e.g. the indicator of a dense set with sparse sections).
- Riemann integration handles unbounded functions and unbounded domains only through improper limits; Lebesgue's theory (module 3.1) removes this awkwardness and gives much stronger Fubini–Tonelli and convergence theorems.

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9. Change of variables

Theorem. Let $U, V \subseteq \mathbb{R}^n$ be open, $g : U \to V$ a C^1 bijection with $\det Dg \neq 0$ everywhere (a C^1 diffeomorphism), and f integrable on V (e.g. continuous with compact support, or on a Jordan measurable compact subset). Then

$$\int_V f(y)\,dy = \int_U f(g(x))\,|\det Dg(x)|\,dx.$$

Proof sketch.

1. *Linear maps:* a linear map T multiplies volumes of rectangles (and hence of Jordan measurable sets) by $|\det T|$. Check it for elementary matrices (row swaps, scaling a coordinate, shears; shears preserve volume by Fubini/Cavalieri) and use multiplicativity of det.
2. *Locally, g is nearly linear:* on a small cube Q around x, $g(Q)$ is squeezed between $Dg(x)\big((1\pm\varepsilon)Q\big)$ shifted to $g(x)$, so $\operatorname{vol} g(Q) \approx |\det Dg(x)| \operatorname{vol} Q$.
3. *Sum over a fine grid* (or, as in Rudin and Spivak, factor g locally into maps that change one coordinate at a time, use a partition of unity, and induct on n using Fubini and 1-variable substitution).

Standard changes.

- Polar: $(r,\theta) \mapsto (r\cos\theta, r\sin\theta)$, $|\det| = r$.
- Spherical: $(\rho,\varphi,\theta) \mapsto (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$, $|\det| = \rho^2\sin\varphi$.
- Gaussian integral: $\Big(\int_{-\infty}^{\infty} e^{-x^2}dx\Big)^2 = \int_{\mathbb{R}^2} e^{-|x|^2} = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = \pi$ (Problem 18 does this carefully).

Polar coordinates fail to be a diffeomorphism on the set $r = 0$ or $\theta \in \{0, 2\pi\}$, but that set has measure zero, so it does not affect integrals.

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10. Differential forms and Stokes' theorem (preview)

Forms on $\mathbb{R}^n$. A *k-form* on an open $U \subseteq \mathbb{R}^n$ is an expression

$$\omega = \sum_{i_1 < \cdots < i_k} a_{i_1 \cdots i_k}(x)\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

with smooth coefficients. Pointwise, it is an alternating k-linear function of tangent vectors. The wedge product is associative, bilinear, and *graded commutative*: $\alpha \wedge \beta = (-1)^{kl} \beta \wedge \alpha$ for a k-form α and l-form β. In particular $dx_i \wedge dx_i = 0$, and $dx_1 \wedge \cdots \wedge dx_n$ is the volume form.

Exterior derivative. $df = \sum_i \partial_i f \, dx_i$ for functions, and $d(a\, dx_I) = da \wedge dx_I$. Properties:

- $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$;
- $d \circ d = 0$ (equality of mixed partials).

In $\mathbb{R}^3$, under the dictionary 0-forms $\leftrightarrow$ functions, 1-forms $\leftrightarrow$ vector fields, 2-forms $\leftrightarrow$ vector fields, 3-forms $\leftrightarrow$ functions, d becomes grad, curl, div, and $d^2 = 0$ becomes $\operatorname{curl} \operatorname{grad} = 0$, $\operatorname{div} \operatorname{curl} = 0$.

Pullback. For a smooth $\phi : V \to U$, ϕ^* substitutes: $\phi^*(a\, dx_{i_1} \wedge \cdots) = (a \circ \phi)\, d\phi_{i_1} \wedge \cdots$. It respects wedge products and commutes with d: $\phi^* d\omega = d\phi^* \omega$. For n-forms, $\phi^*(f\, dy_1 \wedge \cdots \wedge dy_n) = (f \circ \phi) \det(D\phi)\, dx_1 \wedge \cdots \wedge dx_n$: the change of variables formula without absolute values, which is why integrals of forms depend on orientation.

Integration over chains. A *singular k-cube* in U is a smooth map $c : [0,1]^k \to U$, and $\int_c \omega = \int_{[0,1]^k} c^* \omega$. A *$k$-chain* is a formal integer combination of k-cubes. The boundary of the identity cube I^k is

$$\partial I^k = \sum_{i=1}^{k} \sum_{\alpha = 0,1} (-1)^{i+\alpha} I^k_{(i,\alpha)},$$

where $I^k_{(i,\alpha)}$ is the face $x_i = \alpha$; for general c, $\partial c = \sum (-1)^{i+\alpha} c \circ I^k_{(i,\alpha)}$. One checks $\partial \circ \partial = 0$.

Stokes' theorem (on chains). For a $(k-1)$-form ω on U and a k-chain c in U,

$$\int_c d\omega = \int_{\partial c} \omega.$$

Proof idea. By pulling back, it suffices to treat $c = I^k$ and $\omega = a\, dx_1 \wedge \cdots \widehat{dx_i} \cdots \wedge dx_k$; then both sides reduce, via Fubini, to the one-variable fundamental theorem of calculus in the x_i direction.

Special cases: the fundamental theorem of calculus ($k = 1$), Green's theorem, Stokes' theorem for surfaces in $\mathbb{R}^3$, and the divergence theorem. The version for compact oriented manifolds with boundary is proved in module 3.5.

Closed and exact forms. ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$. Exact $\Rightarrow$ closed since $d^2 = 0$. **Poincaré lemma:** on a star-shaped open set, every closed form is exact. On $\mathbb{R}^2 \setminus \{0\}$, the form $\omega = \frac{-y\, dx + x\, dy}{x^2 + y^2}$ ("$d\theta$") is closed, but $\int_{S^1} \omega = 2\pi \neq 0$, so it is not exact (if $\omega = df$, Stokes would give $\int_{S^1} df = 0$). The failure of "closed $\Rightarrow$ exact" measures the topology of the domain: this is de Rham cohomology.

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11. Common pitfalls

1. Interchanging limits (sums, integrals, derivatives) without uniform convergence or a dominating hypothesis.

2. Equicontinuity is about the whole family at once; each function being uniformly continuous is not enough.
3. Stone–Weierstrass needs the constants (or at least non-vanishing at every point) and point separation; the complex version needs conjugation.
4. Existence of partial derivatives does not imply differentiability, or even continuity.
5. The inverse function theorem is local; global injectivity needs extra arguments.
6. Fubini needs integrability on the product.
7. Change of variables uses $|\det Dg|$; integration of forms uses $\det D\phi$ with sign, so orientation matters.

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12. Connections forward

- **3.1 Measure and integration:** Lebesgue integral, dominated convergence (the right interchange theorem), Fubini–Tonelli, change of variables for Lebesgue measure; Riemann integrable ⇔ continuous a.e.
- **3.2 Functional analysis:** $C(K)$ as a Banach space; compact operators via Arzelà–Ascoli; Stone–Weierstrass gives density and separability results; contraction principles in Banach spaces; Riesz representation of $C(K)^*$.
- **3.5 Smooth manifolds:** charts and transition maps are C^∞ diffeomorphisms; regular value theorem from the implicit function theorem; tangent spaces; forms, pullback and Stokes on manifolds; de Rham cohomology.
- **3.4 Algebraic topology:** chains and boundaries ($\partial^2 = 0$) are the start of singular homology; de Rham's theorem relates closed/exact forms to homology.